Technical Report: Algorithmic Optimization of Weekly Training Plans
A Hybrid Constraint-Based Scheduling Approach

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1. Abstract

Designing effective weekly training programs requires balancing multiple competing factors, including exercise selection, fatigue management, time constraints, and biological recovery. In practice, these decisions are often made heuristically. This project formulates workout planning as a formal constrained scheduling problem. We propose a hybrid two-phase approach: a greedy constructive heuristic to build a rapid, feasible baseline, followed by a local search refinement to optimize global workload distribution. By evaluating the algorithms across 30 randomized trials, experimental results demonstrate that the refinement phase improves schedule quality. The system successfully generated a structured 5-day training split that achieved a highly consistent fatigue distribution across the week, demonstrating that computational optimization can effectively support both structural programming goals and estimated workload constraints.

2. Introduction & Motivation

In traditional fitness environments, programming relies heavily on the intuition of coaches and practitioners. However, the selection of individual exercises is highly interdependent. Selecting a neurologically demanding movement like a Barbell Squat on a Monday limits the physiological availability for similar lower-body movements for the subsequent 48 to 72 hours. This creates a cascading decision tree where early choices dictate the feasibility of later programming.

This project addresses this dynamic by modeling weekly planning as a computational problem. Due to the combination of strict limits (time, estimated fatigue) and prerequisite spacing (recovery), this challenge closely mirrors the Resource-Constrained Project Scheduling Problem (RCPSP) widely studied in operations research (Pinedo, 2016). By applying a formal algorithmic framework, this approach aims to reduce heuristic bias and provide a systematic method for balancing weekly workloads.

3. System Architecture and Algorithmic Design

Given the NP-hard nature of resource scheduling, exact optimization methods (like dynamic programming) become computationally intractable as the exercise database grows. Therefore, the optimization pipeline operates in two distinct phases:

- **Constructive Phase (Greedy Initialization):** This phase evaluates feasible exercises and selects the highest locally-scoring move for a given session. To prevent early days from monopolizing high-priority exercises, the system processes days in a non-chronological "spread" sequence (e.g., Monday, Thursday, Tuesday, Friday). This helps distribute high-value compound lifts across the week before constraints lock them out.

- **Refinement Phase (Local Search):** Greedy algorithms frequently become trapped in local optima, producing schedules with uneven daily workloads (Russell & Norvig, 2020). To counter this, the local search engine explores a neighborhood of adjacent schedules via four specific mutations: Insertions, Relocations, Replacements, and Swaps. The algorithm iteratively accepts mutations that minimize the variance of the modeled fatigue score.

4. Constraint Modeling and Domain Knowledge

The core of the scheduler relies on translating qualitative sports science principles into quantitative data structures. Exercises are modeled via JSON schemas, and constraints are enforced through a strict filtering layer:

- **Fatigue Management (Session-RPE Proxy):** Fatigue is abstracted as a simplified integer value representing a bounded resource. While it is a heuristic proxy for complex physiological fatigue, each session is capped at a strict limit (e.g., 25 points) to manage acute modeled workload, drawing upon principles from RPE-based training load models (Impellizzeri et al., 2005).
- **Neurological Recovery Windows:** The system mathematically enforces mandatory spacing between similar movement categories. Heavy compound lifts trigger a 48-to-72 hour lockout for their respective categories to approximate the central nervous system and muscle protein synthesis repair timelines (Haff & Triplett, 2015).
- **Biomechanical Taxonomy:** To ensure structural completeness, the engine seeks to satisfy minimum weekly frequencies across foundational movement patterns: Squat, Hinge, Push, Pull, and Core (Schoenfeld et al., 2016).

5. Methodology & Experimental Setup

To evaluate the system, the algorithms were tested in a controlled environment using a standardized exercise dataset. Because the baseline comparison relies on a Random valid generator, performance was averaged across 30 randomized trials to ensure statistical stability. The evaluation function scores plans based on a weighted sum of six metrics: Coverage Score, Priority Score, Time Utilization, Fatigue Balance (variance penalty), Target Frequency (aiming for a 5-day split), and Hard Constraint Violations.

6. Experimental Results & Output

The experimental run compared the Random baseline, the Greedy baseline, and the fully Refined (Local Search) algorithm. The results are summarized below:

Algorithm	Mean Total Score	Runtime (s)	Modeled Fatigue Balance
Greedy	61.79	~0.0002	0.13
Random	63.80	~0.0017	0.22

Refined	73.80	0.3032	1.00

The data illustrates the utility of the Local Search phase. While the Greedy algorithm executed almost instantaneously, it exhibited poor fatigue distribution (0.13), often front-loading the week and leaving later days sparse. The Refinement phase significantly improved this distribution, achieving a modeled fatigue balance score of 1.0 (representing minimal variance in the assigned training load). The final output produced the following 5-day schedule:

- **Monday & Tuesday (Modeled Fatigue 15, 16):** Heavy lower-body focus (Romanian Deadlift, Back Squat, Trap Bar Deadlift).
- **Wednesday & Saturday (Modeled Fatigue 15, 15):** Upper-body and accessory focus (Bench Press, Barbell Row, Face Pull).
- **Sunday (Modeled Fatigue 15):** Secondary lower-body compound session (Conventional Deadlift, Bulgarian Split Squat).

7. Discussion of Engineering Challenges

Translating physiological concepts into strict constraints required several iterations of tuning. Key engineering challenges included:

- **The Meaningful Session Threshold:** Initially, the local search engine aggressively deleted exercises to balance the fatigue score, resulting in fragmented 1-exercise "micro-sessions." Implementing a strict lower bound (minimum 10 minutes, minimum 2 exercises) was necessary to ensure all active days represented a sufficient training stimulus.
- **Fatigue Multipliers vs. Priority Weights:** Balancing the fatigue penalty against the inherent priority score of an exercise required significant heuristic tuning. If the fatigue penalty was too aggressive, the system ignored compound lifts entirely; if too lenient, it consistently hit daily caps with lower-priority selections.

8. Reproducibility

To ensure results can be independently verified, the project utilizes standard Python 3.10+ environments, fixed JSON schemas for both exercises and client requests, and a comprehensive pytest suite. Randomized generations were locked to fixed seeds (seed=42) during testing to allow for deterministic validation.

9. Future Work

The current system optimizes for 'Flat Fatigue,' seeking to equalize daily modeled RPE. Future iterations will transition the data model to support Undulating Periodization (Fleck & Kraemer, 2014). This will involve modifying the daily fatigue cap from a static integer into an oscillating array (e.g., Heavy, Light, Medium days) to better represent advanced, multi-week macrocycle planning.

10. Conclusion

This project demonstrates that weekly training program design can be formulated as a constrained optimization problem. The experimental results indicate that while greedy heuristics are effective for rapid initialization, the application of local search refinement is essential for escaping local optima and achieving a more uniform workload distribution. This hybrid architecture provides an extensible, interdisciplinary framework for generating constraint-compliant training plans, while acknowledging the inherent simplifications required to model physiological fatigue computationally.

11. References

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